

both domestic and foreign investments in the nuclear power sector.

During the meeting, the industry raised the question of high operation and maintenance (O&M) costs and challenges in raising money from the market. However, they were told that the government would sign the guarantee to facilitate capital lending.

As per the RFP criteria, O&M expenditure is around ₹75.65 lakh per MW electric/annum for the year 2025-26, with an annual escalation of 6% per annum for every subsequent year and water charges, cess, levies and insurance charges at actuals.

However, NPCIL says that the O&M and fuel costs of nuclear power are lower, and the significant component is the investment cost. “Our latest 700 MW PHWRs have been set up at a cost of about ₹16 crore/MW, which is highly competitive on a global scale. However, the actual tariffs of nuclear power plants to be set up in future would depend on factors like cost of financing, gestation period, and extent of localisation in respect of imported technologies,” it said.

Nikhil Dhingra, CEO, Acme Solar, highlights that for any of this to take off, the key thing should be the cost competitiveness of this source of power. “The cost competitiveness needs to be there, and the commercial construct needs to be solid in terms of the economics and the safety of the investment, and also in terms of the viability of the investment,” he tells *BT*.

The NPCIL, in its RFP, has said that BSR is for captive use, but the industry can sell the excess power at a tariff decided by them. Dhingra suggests that the power sale price should be disclosed so that investors can assess the potential return on their investment.

According to analysts, project risks are higher in case of nuclear power plants vis-a-vis other power plants due to high upfront capital costs (₹15-17 crore per MW compared to less than ₹10 crore per MW in other types of power plants), as well as the higher gestation period (five-six years v/s <four years).

At the same time, the operational phase risks are reduced because of more reliable plant load factors than when using renewable energy and cheaper power generation than with thermal power. Additionally, with the asset life being 60 years, this can spread the high upfront capital expenditure over the project lifespan.

Sehul Bhatt, Director- Research, Crisil Intelligence, says that the introduction of a public-private partnership model, coupled with the allocation of ₹20,000 crore for indigenous reactors, should attract private investments, improve capital deployment and accelerate capacity addition. “To further develop this segment, the

STATES' CAPACITY

- ▶ Rajasthan will soon pip Tamil Nadu as the state with the highest nuclear capacity
- ▶ More states to add nuclear power capacity

STATES	NUMBER	CAPACITY (MW)
Uttar Pradesh	2	440
Karnataka	4	880
Maharashtra	4	1,400
Gujarat	4	1,840
Rajasthan	7	1,880
Tamil Nadu	4	2,440
Total	25	8,880

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industry should ensure reliable supply chains for the fuel (a rare mineral), implement stringent safety protocols and effective waste disposal mechanisms, and use small reactors for commercial applications,” Bhatt tells *BT*.

The industry is expecting more clarity on costs, tariffs and operational parts in the coming months.

LEGAL AMENDMENTS

Nuclear power in India is regulated under the Atomic Energy Act and the Civil Liability for Nuclear Damage Act. To facilitate private participation in the nuclear sector, amendments to these legislations is being undertaken. Presently, all nuclear power reactors are un-